

NEWINGTON COLLEGE



2014 Assessment 2

Year 11 Mathematics (2 Unit)

General Instructions:

- Date of task - Wednesday 28th May (Wk 16B)
- Working time – 45 mins
- Weighting - 15%
- Board-approved calculators may be used.
- Attempt all questions, start each question in a new booklet.
- Show all relevant mathematical reasoning and/or calculations.

Total marks – 39

Question / Outcome	Real Functions	Trigonometry
Q1 Multiple choice	/2	/3
Q2	/3	/8
Q3	/7	/4
Q4	/7	/5
Total	/19	/20

Outcomes to be assessed:

- P3** performs routine arithmetic and algebraic manipulation involving surds, simple rational expressions and trigonometric identities
- P4** chooses and applies appropriate arithmetic, algebraic, graphical, trigonometric and geometric techniques

Question 1 (5 marks) - Answer on the multiple choice answer sheet.

(a) The domain of the function $y = \sqrt{4 - x^2}$ is defined as

A $-4 \leq y \leq 4$

B $-2 \leq x \leq 2$

C $-2 \leq y \leq 2$

D $-4 \leq x \leq 4$

(b) Given $\cos \theta = -\frac{3}{5}$ and $\sin \theta < 0$, the ratio for $\tan \theta =$

A $\frac{4}{5}$

B $-\frac{4}{3}$

C $-\frac{4}{5}$

D $\frac{4}{3}$

(c) The exact value of $\sin 225^\circ$ is

A $-\frac{1}{\sqrt{2}}$

B $\frac{1}{\sqrt{2}}$

C $\sqrt{2}$

D $-\sqrt{2}$

(d) The coordinates of the centre and the length of the radius of the circle

$$(x - 4)^2 + y^2 = 36 \text{ are}$$

A Centre $(-4,0)$, radius = 6

B Centre $(4,1)$, radius = 6

C Centre $(4,0)$, radius = 36

D Centre $(4,0)$, radius = 6

(e) The graph of $y = \tan x$ in the domain $0^\circ \leq x \leq 360^\circ$ has asymptotes at

A $x = 90^\circ$ and $x = 270^\circ$

B $x = 180^\circ$

C $x = 360^\circ$

D $x = 0^\circ$ and $x = 180^\circ$

End of Question 1

Question 2 (11 marks) - Start a new booklet

(a) What is the vertex of the parabola $y = -x^2 + 5$? 1

(b) Find the exact value of $\frac{2 \tan 30^\circ}{\cos 45^\circ}$, leaving your answer with a rational denominator. 2

(c) Solve the following equations in the domain $0^\circ \leq \theta \leq 360^\circ$ (to the nearest minute) 4

(i) $\sin \theta = \frac{1}{4}$

(ii) $2 \tan \theta + 1 = 0$

(d) Given 2

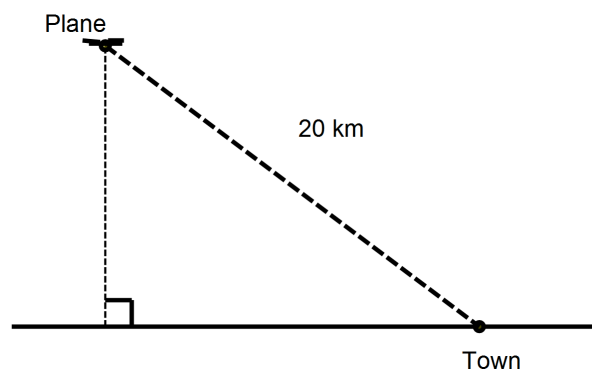
$$f(x) = \begin{cases} 1 - x & \text{for } x \leq 0 \\ 1 - x^2 & \text{for } 0 < x < 1 \\ |x - 1| & \text{for } x \geq 1 \end{cases}$$

evaluate $f\left(\frac{1}{2}\right) - 2f(-3) + f\left(2\frac{1}{2}\right)$

(e) A plane is flying horizontally at a steady speed, 2 km above the ground. 2

A town can be seen from the plane 20 km away.

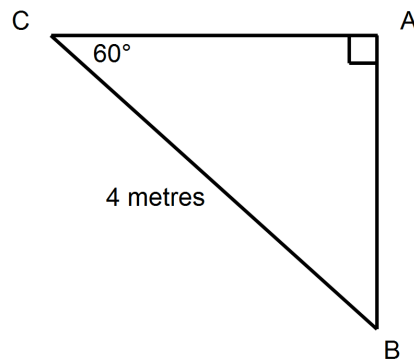
Find the angle of depression of the town from the plane, correct to the nearest degree.



End of Question 2

Question 3 (11 marks) – Start in a new booklet

- (a) Find the exact length of the side AB in the triangle ABC

2

- (b) Find the exact value of $\operatorname{cosec}(-120^\circ)$

2

Leave your answer with a rational denominator.

- (c) If $f(x) = 4x^2 - 4x - 7$

- (i) find $f(k-1)$ in simplest form

2

- (ii) find the value(s) of k if $f(k-1) = 1$

2

- (d) Sketch the intersection of the regions

3

$$y \geq x^2 + 4 \quad \text{and} \quad x + y < 6$$

on your own number plane. All intercepts must be shown.

End of Question 3

Question 4 (12 marks) – Start in a new booklet

- (a) For the following functions, sketch the graph, showing all intercepts and asymptotes if they exist 6

(i) $y = \frac{-2}{x-1}$

(ii) $y = (x+2)^3$

(iii) $y = 2|x| - 4$

- (b) State the range of $y = 2^{-x} + 1$ 1

- (c) Maya drove 12 kilometres from home(**H**) to the beach(**B**) on a bearing of 254° to pick up her children. She then drove to the supermarket(**S**), which has a bearing of 344° from her home. The distance between the supermarket and her home is 4.5 kilometres.

- (i) Draw a neat diagram representing this information 1

- (ii) Show that the $\angle SHB$ is 90° 1

- (iii) Find the bearing of the supermarket **from** the beach. 3

Round your answer to the nearest degree.

End of Examination

Student Name:..... Number: Teacher:.....

Year 11 Mathematics Q1 – Multiple Choice Answer Sheet

Completely fill the response oval representing the most correct answer.

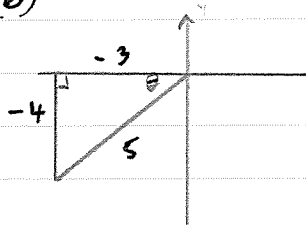
- | | | | | | | | | |
|---|---|-----------------------|---|-----------------------|---|-----------------------|---|-----------------------|
| 1 | A | ☐ | B | ☐ | C | ☐ | D | ☐ |
| 2 | A | ☐ | B | ☐ | C | ☐ | D | ☐ |
| 3 | A | ☐ | B | ☐ | C | ☐ | D | ☐ |
| 4 | A | ☐ | B | ☐ | C | ☐ | D | ☐ |
| 5 | A | ☐ | B | ☐ | C | ☐ | D | ☐ |

QUESTION 1 - multiple choice
(5 MARKS)

(a) domain

$$-2 \leq x \leq 2 \quad \underline{B}$$

(b)



$$\begin{aligned} \tan \theta &= \frac{-4}{-3} \\ &= \frac{4}{3} \end{aligned}$$

D

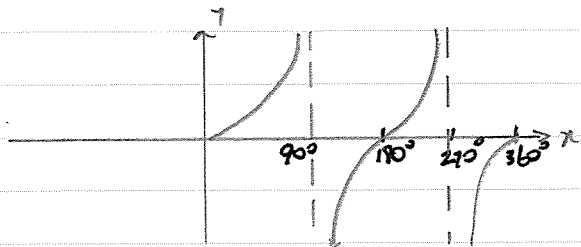
$$\begin{aligned} (c) \sin 225^\circ &= \sin(180^\circ + 45^\circ) \\ &= -\sin 45^\circ \\ &= -\frac{1}{\sqrt{2}} \end{aligned}$$

A

(d) centre (4, 0)
radius 6

D

(e)



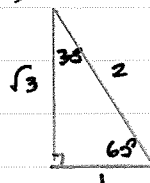
Asymptotes at $x = 90^\circ$
 $x = 270^\circ$

A

QUESTION 2 (11 MARKS)

(a) vertex (0, 5) 1

(b)



$$\frac{2 \times \frac{1}{\sqrt{3}}}{\frac{1}{\sqrt{2}}} = \frac{\frac{2}{\sqrt{3}}}{\frac{1}{\sqrt{2}}} \quad \underline{1}$$

$$= \frac{2\sqrt{2}}{\sqrt{3}} \times \frac{\sqrt{3}}{\sqrt{3}}$$

$$= \frac{2\sqrt{6}}{3} \quad \underline{1}$$

(c) (i) $\theta = 14^\circ 29'$ 1st QUAD 1

$\theta = 165^\circ 31'$ 2nd QUAD 1

(ii) $2 \tan \theta = -1$

$$\tan \theta = -\frac{1}{2}$$

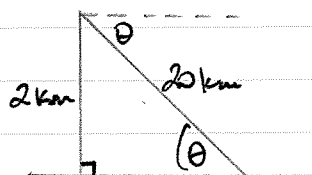
basic angle: $\theta = 26^\circ 34'$

2nd quad: $\theta = 153^\circ 26'$ 1

4th quad: $\theta = 333^\circ 26'$ 1

$$\begin{aligned} (d) \left(1 - \left(\frac{1}{2}\right)^2\right) - 2 \times (1 - -3) + \left|2\frac{1}{2} - 1\right| \\ = \frac{3}{4} - 8 + \frac{1}{2} = -5\frac{3}{4} \quad \underline{1} \end{aligned}$$

(e)



$$\sin \theta = \frac{2}{20} \quad \underline{1}$$

$$\begin{aligned} \theta &= 5^\circ 44' 2'' \\ &= 6^\circ \text{ (to the nearest degree)} \quad \underline{1} \end{aligned}$$

QUESTION 3 (11 MARKS)

(a) $\sin 60^\circ = \frac{AB}{4}$

$AB = 4 \times \frac{\sqrt{3}}{2}$ ✓

$= 2\sqrt{3} \text{ metres}$ ✓

(b) $\operatorname{cosec}(-120^\circ)$

$= \frac{1}{\sin(-120^\circ)}$

$= \frac{1}{\sin 240^\circ}$ ✓

$= \frac{1}{-\sin 60^\circ}$

$= -\frac{1}{\sqrt{3}/2} = -\frac{2}{\sqrt{3}}$ ✓

(c) (i) $f(k-1)$ ✓

$= 4(k-1)^2 - 4(k-1) - 7$

$= 4(k^2 - 2k + 1) - 4k + 4 - 7$

$= 4k^2 - 8k + 4 - 4k - 3$

$= 4k^2 - 12k + 1$ ✓

(ii) If $f(k-1) = 1$,

$4k^2 - 12k + 1 = 1$

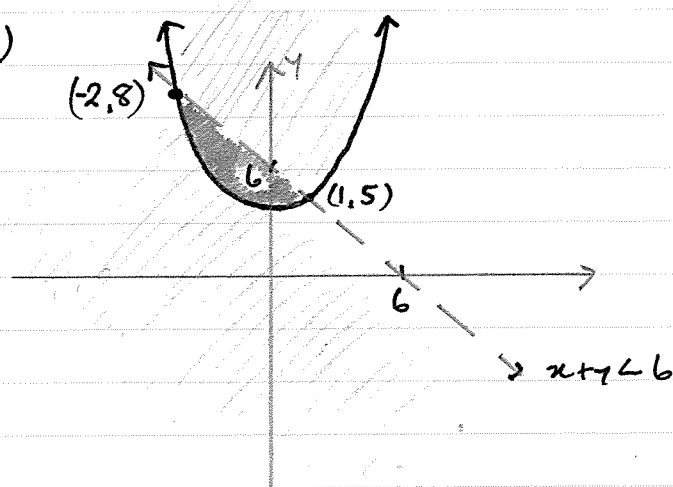
$4k^2 - 12k = 0$

$4k(k-3) = 0$

$\therefore k = 0 \text{ or } k = 3$.

✓ ✓

(d)



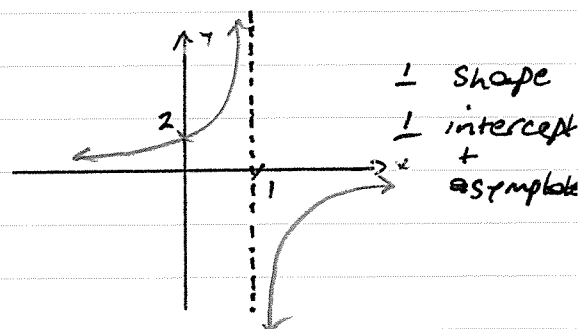
✓ for $x + y < 6$

✓ for $y > x^2 + 4x + 6$

✓ for correct shaded region

QUESTION 4 (12 MARKS)

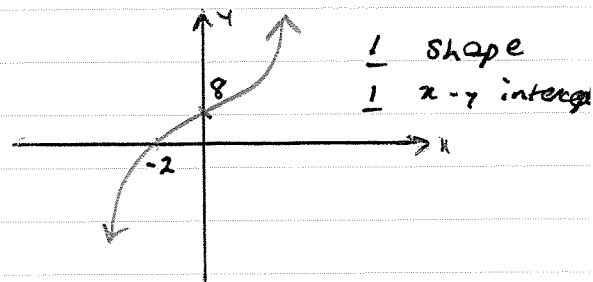
(a) (i)



✓ shape

✓ intercept + asymptote

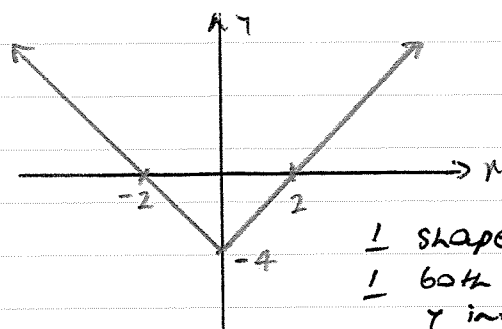
(ii)



✓ shape

✓ $x - y$ intercept

(iii)



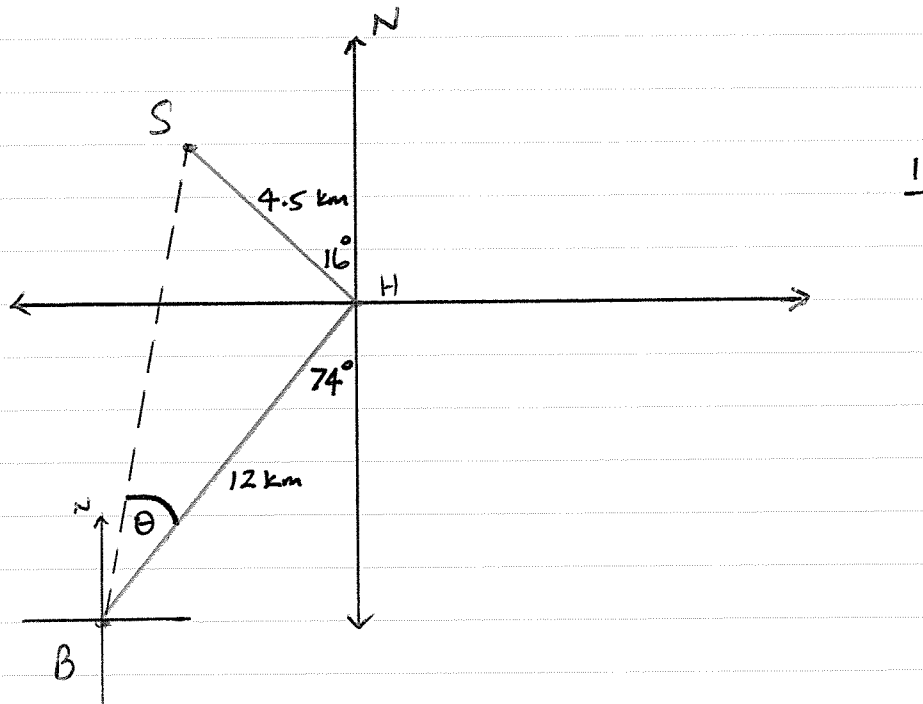
✓ shape

✓ both x and y intercepts

QUESTION 4 cont'd

(b) Range $y > 1$ 1

(c) (i)



(ii) $\angle SHN^{\text{th Pole}} = 16^\circ$

$\angle BHN^{\text{th Pole}} = 74^\circ$

$\angle SHB = 180^\circ - 74^\circ - 16^\circ = 90^\circ$ 1

(iii) From the diagram

Find θ : $\tan \theta = \frac{4.5}{12}$ 1

$\theta = 20^\circ 33' 22''$ 1

$\angle NBH = 74^\circ$ (alternate angle)

Bearing : $74^\circ - 20^\circ 33' 22''$
 $= 53^\circ 26' 38''$
 $= \underline{053^\circ}$ to the nearest degree 1